Numerical Simulation Methods And Their Applications

Monte Carlo Methods and Applications in Particle-based Systems

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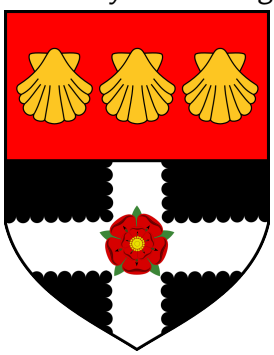

April 24, 2026

Plain language summary

This project concerns working with a computer simulation of a particle-based system, particularly using numerical techniques known as Monte Carlo simulations.

It starts with an introduction to the history of these methods, touching upon the Manhattan Project, then focusing on pseudo-random number generators, sampling methodologies, and their applications in statistical mechanics. The project also explores the Metropolis method and its application to a Lennard-Jones liquid. Finally, it concludes with findings of a simulation created from individually developed C++ code.

These findings consist of elements such as total energy within the system, and values derived from numerical analysis of the particles, such as (but not limited to) the radial distribution and diffusion coefficients.

Declaration

I certify that this is my own work and that use of material from other sources has been properly and fully acknowledged in the text. I have read the University's definition of plagiarism and the department's advice on good academic practice. I understand that the consequence of committing plagiarism, if proven and in the absence of mitigating circumstances may include failure in the Year or Part or removal from the membership of the University. I also certify that neither this piece of work, nor any part of it, has been submitted in connection with another assessment.

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1 Development Of Computer Simulation Methods

The first concepts of Monte Carlo simulations can be found in the 18th century, contained within a report found in the 1733 edition of 'Histoire de L'Académie Royale des Sciences'. This report - created by Georges Louis Leclerc, Comte de Buffon - makes mention of an approximation of pi by throwing needles or rods atop a floor of boards, measuring the fractions that fell on a single board. Of course, this can be recreated in the modern day by a simulation of approximating pi using many points (see Appendices A and B). When initially proposed in this report, the field of Monte Carlo methods had not been distinctly named or partitioned (Landau & Binder 2021).

1.1 The Manhattan Project

1.1.1 Background Context

Before understanding Monte Carlo Methods we are bound to mention **The Manhattan Project**. The top-secret research and development program was the first to produce Nuclear Weapons, and was a key playing card in the context of the greater geopolitical climate: World War II.

Funded by nations such as the UK, Canada and, majorly, the USA, the Manhattan Project is often cited as a turning point in human history. One of the sites of the project was Los Alamos, New Mexico - tied intimately with the birth of Monte Carlo methods. Disregarding moral arguments concerning atomic weaponry, a large factor in the awe of the project was the pure advancement in scientific knowledgement - building upon previous discoveries of fission.

1.1.2 Stanisław Ulam

Stanisław Ulam was prominent in contributing to the founding of Monte Carlo Methods as a designated methodology, during and after his involvement in The Manhattan Project. Famously, Ulam thought up the concept while recovering in a hospital from surgery. He had been playing solitaire to pass the time and wondered whether the probability of success could be estimated statistically. Rather than approaching the problem from a combinatorial approach (which he had already tried and found unsuccessful) he begun to think more 'abstractly' - he had realised that he could simply observe successive games and note outcomes. Crucially for Stanisław Ulam, this could be extended to many simulated trials with the recent advent of computers - which would yield increasingly accurate estimates (Eckhardt 1987).

1.1.3 Subsequent Developments

The link between Ulam's proposed ideas and the following adoption of the statistical approach was a man named John Von Neumann. Perhaps most well-known for his invention of the aptly named Von Neumann architecture (in which programs are stored within a computer much like the data these programs use), he provided the link to the ENIAC - the world's first programmable, electronic digital computer. Von Neumann's computations, refined together with colleagues at Los Alamos laboratory, became the first calculations run on the ENIAC after it had been upgraded to a stored-program paradigm. Nowadays, the computer industry has grown exponentially, and though statistical sampling had already been used in the past, Ulam's thoughts paved the way for the Monte Carlo method to dominate complex statistical problems (Summerscales 2023).

2 Monte Carlo Methods

2.1 Pseudo-Random Number Generators

Random number generation is an important part of Monte Carlo methods. There are multiple different methods to obtaining random numbers, but the algorithms we are focusing on within this report are called **Pseudo-Random Number Generators** (PRNGs). While they do not generate 'actual' random numbers, what they generate is sufficient enough in variability that it can be used in scenarios where random numbers are needed.

The way that they achieve this is through the use of mathematical formulae, varying in levels of complexity. The algorithms can get increasingly elaborate, yet the simplest can consist of only a few lines of code utilising basic mathematical arithmetic. Sequences are generated which attempt to approximate random numbers, based on an initial starting *seed*, and these sequences will eventually repeat after their *period*. For satisfactory generators, these periods will be sufficiently long enough that the repititions are often unreached by realistic samples (Gentle 2003).

While humans can not predict the next number in the sequences that are generated, this is not perfectly computationally secure. This is because if the initial *seed state* is known, an identical sequence can be reproduced. Therefore, the sequences are deterministic and not as unpredictable as systems with true chaos built into them - such as radioactive atom decay, or, famously, *Cloudflare*'s wall of lava lamps (Liebow-Feeser 2017).

2.1.1 The Linear Congruential Generator (LCG)

One such example of a PRNG, albeit possibly the simplest one, is the Linear Congruential Generator given in the following form:

$$x_i \equiv (ax_{i-1} + c) \bmod m$$

As can be seen, the equation consists of a simple linear function followed by a modular reduction. Although this may not produce sufficiently long streams of random numbers, and the numbers may not appear to be very random themselves, the Linear Congruential Generator and its alternative forms create the basis of many more powerful PRNGs. These alternative forms are built by modifying certain elements, for example setting $c = 0$ (yielding something known as the Multiplicative Congruential Generator)

Here,

- a is called the 'multiplier'
- c is called the 'increment'
- m is called the 'modulus'.

The seed for this generator is the inital value supplied, which is to say x_0. Due to the modular reduction that occurs, there can only be m possible values that x can take; The maximum period of the LCG is, by definition, m (Gentle 2003).

2.1.2 Linear Congruential Generator Example

Following is an example of a LCG displayed in a 'truth table', allowing the process to be broken down into small, simple, and visible steps. Traditionally, values of m are usually a power of 2 as this is easier for the computer to use during calculation. Due to this, I have chosen $m = 2^5 = 32$ which is smaller than a typical usecase. The other values that this LCG will take are $a = 5,\ c = 7,\ X_0 = 19$, which have been arbitrarily selected.

Linear Congruential Generator Example

aX_n	$+ c$	$(\mathrm{mod}\ m)$	$= X_{n+1}$
$(5 * 19) = 95$	$(95 + 7) = 102$	$102\ (\mathrm{mod}\ 32)$	$= 6$
$(5 * 6) = 30$	$(30 + 7) = 37$	$37\ (\mathrm{mod}\ 32)$	$= 5$
$(5 * 5) = 25$	$(25 + 7) = 32$	$32\ (\mathrm{mod}\ 32)$	$= 0$
$(5 * 0) = 0$	$(0 + 7) = 7$	$7\ (\mathrm{mod}\ 32)$	$= 7$
$(5 * 7) = 35$	$(35 + 7) = 42$	$42\ (\mathrm{mod}\ 32)$	$= 10$
$(5 * 10) = 50$	$(50 + 7) = 57$	$57\ (\mathrm{mod}\ 32)$	$= 25$
$(5 * 25) = 125$	$(125 + 7) = 132$	$132\ (\mathrm{mod}\ 32)$	$= 4$

From the table, we can see that for the Linear Congruential Generator $5X_n + 7\ (\mathrm{mod}\ 2^5)$, the starting seed of $X_0 = 19$ provides a sequence of: $6, 5, 0, 7, 10, 25, 4 \cdots$.

Following this, the sequence continues until it starts to repeat, having a period of 32 (the value of m, as stated earlier). The full sequence in its entirety is:

$$19, 6, 5, 0, 7, 10, 25, 4, 27, 14, 13, 8, 15, 18, 1, 12, 3, 22, 21, 16, 23, 26, 9, 20, 11, 30, 29, 24, 31, 2, 17, 28, 19, \ldots$$

2.1.3 The Mersenne Twister

The Mersenne Twister is a more complex, modern PRNG developed in 1997 (Matsumoto & Nishimura 1998).

It derives its name from a prime number known as a Mersenne Prime, which contributes to its period. As its period is sufficiently large, the algorithm is highly optimised, and the numbers generated have low correlations, the Mersenne Twister is a prominent PRNG that is used across industry. Most notably, it is used in scientific simulations (Monte Carlo methods, as discussed) but also within Machine Learning, statistical sampling and also video game graphics (Samadov 2024).

2.2 Sampling Methodology

By 'Sampling Methodology', we refer to the way in which Monte Carlo methods are used to sample data points (selections to be respresented within subsequent analysis). Depending on the context and scenario, different qualities may be prioritised, and this is reflected in the different ways that sampling can occur (Brownlee 2019).

The 'Law of Large Numbers' plays a significant part in these simulations as that allows the sampling that occurs to form a relvant distribution, despite the initial data being random.

2.2.1 Simple Sampling

Simple sampling, also called a 'crude monte carlo', is the most direct and rudimentary method of using monte carlo sampling techniques. This entails the collection of completely random observations of a random variable. The aformentioned reliance on the law of large numbers leads to a large enough sample size so that an estimate of the expected value ($E[X]$) can be found (*Simple Sampling* 2010).

The way that simple sampling works is based upon the integral of a collected distribution. If a one-dimensional integral I, where $I = \int_a^b \, dx \; f(x)$, were to be re-written as:

$$I = (b - a) \; \langle \; f(x) \; \rangle,$$

this would represent the unweighted average of $f(x)$ over the interval $[a, b]$. As this average is evaluated, larger and larger distributions tending to infinity yield a correct value of I (Frenkel & Smit 1996).

An example of this 'crude monte carlo' method can be found in the approximation of Pi mentioned earlier (see Appendices A and B), where uniform and random coordinates are generated within the given square. This results in a calculated average for the value of Pi. This sampling methodology can fall short due to its reliance on randomness; If random values fell unusually within neglibible or heavily influencing areas, the final results of a monte carlo sample can be innacurate - rendering the process inefficient.

2.2.2 Importance

Like Simple sampling, Importance sampling is a method of sampling for data values over an interval, however what sets Importance sampling apart is the addition of weighted 'importance' to certain selected values (hence the name given to the technique). How this is achieved is through the introduction of a new weight term This means that variation is reduced to either insignificant levels, or at best zero, in areas where the distribution may not be relevant to draw samples from.

2.3 Statistical Mechanics

2.4 The Metropolis Method

3 The Lennard-Jones Potential

3.1 A Description

The Lennard-Jones potential describes the potential energy of two non-bonded particles based on their distance of separation. The equation accounts for both attractive forces and repulsive forces.

The equation is given as follows

$$V(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

where:

- V indicates the intermolecular potential between particles,
- ε is the well depth: the maximum strength of attractive force and minimum energy value
- σ is referred to as the **'van der Waals'** radius, and is a measure of how close two nonbonding particles can get. It measures the distance at which the intermolecular potential between the two particles is zero.
- r indicates distance of separation between both particles, measured from the center of each.

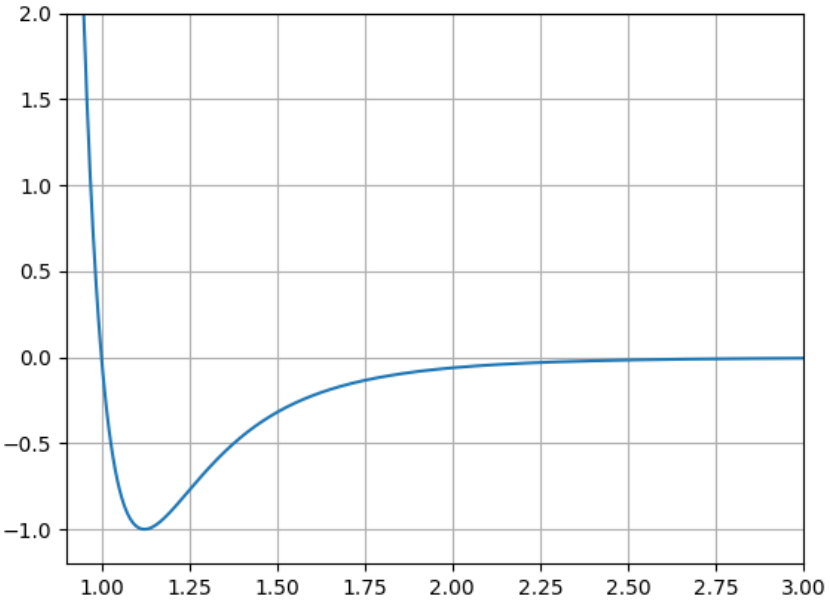
The σ/r^{12} term models strong, short-range repulsion and the σ/r^{6} term models long-range van der Waals attraction.

Sometimes the potential can be rearranged and expressed alternatively as

$$V(r) = \frac{A}{r^{12}} - \frac{B}{r^{6}}$$

in which A, B are equal to $4\varepsilon\sigma^{12}, 4\varepsilon\sigma^{6}$ respectively (*Lennard-Jones Potential* 2025).

3.1.1 Plotting The Potential



3.1.2 Applications in Modelling Simple Liquids

When modelling simple liquid simulations, the Lennard-Jones potential is relevant because

3.2 Periodic Boundary Conditions

4 Monte Carlo Simulation: Code And Development

4.1 Steps Toward Implementation

4.1.1 Drawing Together The Theory

4.1.2 Initial Configuration

4.1.3 Trajectory Generation Loops

4.1.4 Post-Simulation Data Analysis

4.2 Code Origin and Inspiration

My code was built on a framework of **Fortran** code from 'Understanding Molecular Simulation' by Daan Frenkel and Berend Smit (Frenkel & Smit 1996). Another Monte Carlo simulation of the Lennard-Jones model (Kelter et al. 2015), packaged within a library known as NetLogo (Wilensky 1999), also provided much guidance and inspiration. These were adapted and rewritten from scratch, by me, into **C++**. My program can be found in the appendix of this report (see Appendix C).

5 Application of Code to a Lennard-Jones liquid

5.1 Total Energy in Plots

5.2 Numerical Analysis

5.2.1 Radial Distribution Function

5.2.2 Mean-Squared Displacements

5.2.3 Diffusion Coefficients

6 Conclusions, Findings, Prospectives

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A Approximation of Pi in Python, Using Monte Carlo Techniques

ApproximationOfPi/src/python/SimpleImplementation.py

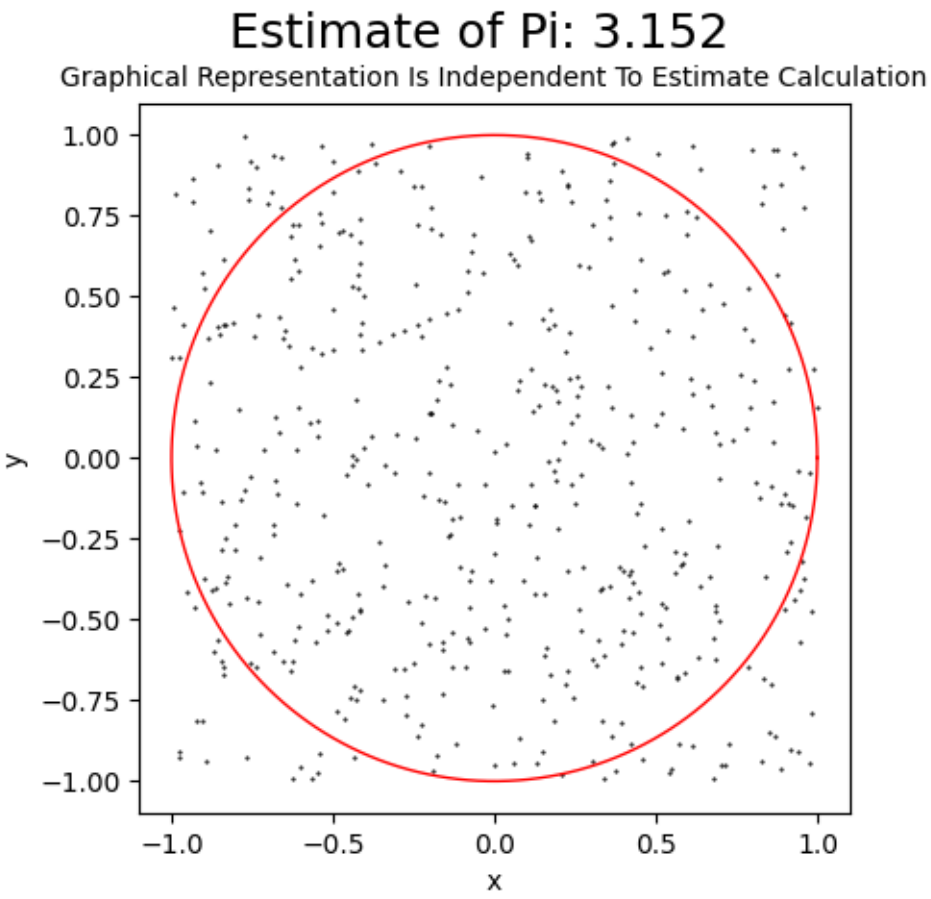
```
#! python3
"""
Simple implementation of approximation of pi.

Steps:
Generate random x, y inside centre of square of side 2
Repeat steps for a large number of points
Calculate ratio of points
Multiply by 4 (pi / 4)
"""

import numpy as np

def estimatePi(samples):
    insideCircle = 0
    for i in range(samples):
        x = np.random.uniform(-1, 1)
        y = np.random.uniform(-1, 1)

        if (x**2 + y**2) <= 1:
            insideCircle += 1
    ratio = insideCircle / samples
    return (ratio * 4)

if __name__ == "__main__":
    print(f"Estimate with 1 Sample:\n{estimatePi(1)}\n")
    print(f"Estimate with 10 Sample:\n{estimatePi(10)}\n")
    print(f"Estimate with 50 Sample:\n{estimatePi(50)}\n")
    print(f"Estimate with 100 Sample:\n{estimatePi(100)}\n")
    print(f"Estimate with 1000 Sample:\n{estimatePi(1_000)}\n")
    print(f"Estimate with 10,000 Sample:\n{estimatePi(10_000)}\n")
    print(f"Estimate with 1,000,000 Sample:\n{estimatePi(1_000_000)}\n")
```

ApproximationOfPi/src/python/GraphicalRepresentation.py

```python
#! python3
"""
Graphical representation of approximation of Pi using Monte Carlo methods.
"""
import matplotlib.pyplot as plt
import numpy as np

from SimpleImplementation import estimatePi

def main():
    samples = int(input("Enter Intended Samples:\n"))

    x = np.random.uniform(-1, 1, samples)
    y = np.random.uniform(-1, 1, samples)

    plt.plot(x, y, ".k", markersize = 1)

    t = np.linspace(0, 2 * np.pi, 400)
    xc = np.cos(t)
    yc = np.sin(t)
    plt.plot(xc, yc, 'r-', linewidth=1)
    plt.gca().set_aspect('equal', adjustable='box')

    plt.xlim(-1.1, 1.1)
    plt.ylim(-1.1, 1.1)
    plt.xlabel("x")
    plt.ylabel("y")

    plt.suptitle(f"Estimate of Pi: {estimatePi(samples)}", fontsize = 18)
    plt.title("Graphical Representation Is Independent To Estimate Calculation",
        fontsize = 10)
    plt.show()

if __name__ == "__main__":
    main()
```

B Approximation of Pi in Python, Code Outputs

```
Estimate with 1 Sample:
4.0

Estimate with 10 Sample:
3.2

Estimate with 50 Sample:
2.96

Estimate with 100 Sample:
3.04

Estimate with 1000 Sample:
3.104

Estimate with 10,000 Sample:
3.1376

Estimate with 1,000,000 Sample:
3.143152
```



C Monte Carlo C++ Code

ParticleSimulation/src/main.cpp

```cpp
#include <iostream>


int initialisation() {

    // Particle Placement


    return 0;
}

void mcmove() {

    // Particle Displacement

}


int main() {

    // Metropolis Algorithm

    double Temp;
    int NumParticles;
    double Density;
    int Cycles;

    initialisation();


    return 0;
}
```